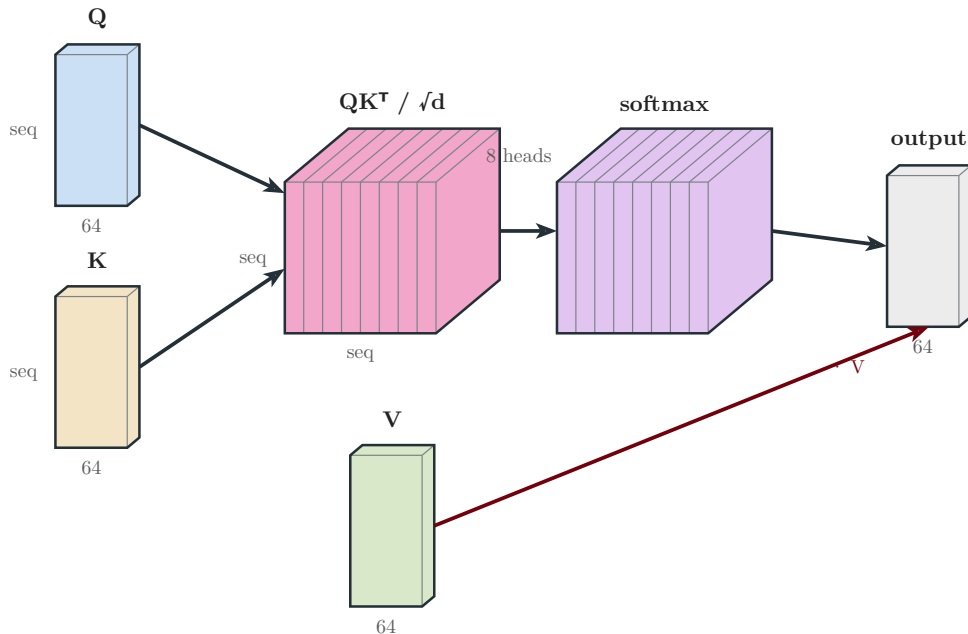


# Multi-head self-attention block

$H = 8$  heads ·  $d_k = d_v = 64$  · scaled dot-product attention  $\text{softmax}(QK^\top / \sqrt{d_k})V$  per head



**Q**, **K**, **V** are linear projections of the input sequence ( $\text{seq} \times d_k$ ). The score tensor  $QK^\top / \sqrt{d_k}$  is a  $\text{seq} \times \text{seq} \times 8$  cube — one  $\text{seq} \times \text{seq}$  attention map per head (the depth seams). After softmax, each head's weights are applied to **V**; the 8 head outputs are concatenated and linearly transformed to the block output.